

Qualifying the 737 CLAUDENet v2 DR9-Sweep Lens Candidates: Expanded Crossmatch and Dual Independent Visual Vetting

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Claude (Anthropic)

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1 Qualifying the 737 ClaudeNet v2 sweep candidates

A campaign to test whether any of the 737 new-and-unseen DR9-sweep candidates (group-conformal $\text{FDR} \leq 0.05$) could be genuinely undiscovered strong lenses, via an expanded crossmatch (SIMBAD + DES/KiDS VizieR; NED/SuGOHI unreachable) and two methodologically-independent visual passes: a structured Huang-VI rubric pass with an adversarial skeptic re-check, and the independent **lensjudge** harness (factored multiagent, Claude-opus). *Qualified* = still NEW after the crossmatch AND graded A/B by BOTH passes.

Result. Of 737 candidates, 136 (18%) coincide with a known lens or lens-candidate (92 DES, plus KiDS and SIMBAD lens-types); **601** remain NEW. *None* of the 601 new candidates is graded A/B by either grader’s strongest pass, so **qualified** = 0. This is not the graders rejecting everything: together they graded 5 candidates $\geq \text{B}$, and **all 5 are already-catalogued lenses** (4 sub-arcsecond DES matches) — the consensus re-discovered real lenses and found nothing comparable among the new ones.

Lenses the campaign re-discovered (already catalogued; the validation):

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Figure 1: s_399907_3947: RA=217.47836, DEC=12.04329, $p_{\text{final}}=0.716$; status KNOWN_REMOTE (nearest curated at $6464.70''$); visual C, lensjudge B. Full | zoom | residual.



Figure 2: s_137518_4818: RA=31.35877, DEC=-35.66318, $p_{\text{final}}=0.935$; status KNOWN_LOCAL (nearest des_jacobs2019 at $0.14''$); visual B, lensjudge B. Full | zoom | residual.



Figure 3: s_240846_692: RA=30.28319, DEC=-15.85474, $p_{\text{final}}=0.898$; status KNOWN_LOCAL (nearest des_jacobs2019 at $0.05''$); visual C, lensjudge B. Full | zoom | residual.



Figure 4: s_137502_4157: RA=26.44498, DEC=-35.69096, $p_{\text{final}}=0.892$; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.19''); visual C, lensjudge B. Full | zoom | residual.

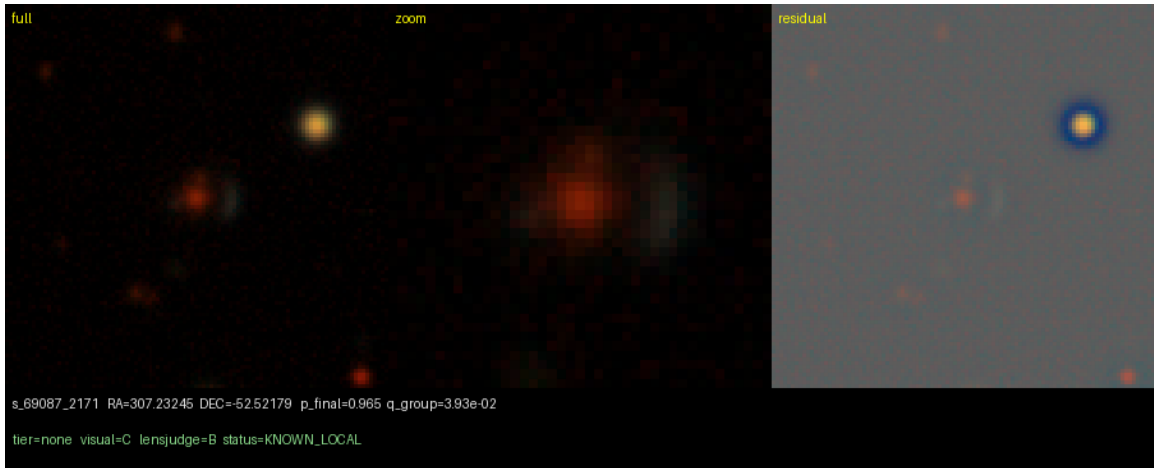


Figure 5: s_69087_2171: RA=307.23245, DEC=-52.52179, $p_{\text{final}}=0.965$; status KNOWN_LOCAL (nearest des_jacobs2019 at 0.09''); visual C, lensjudge B. Full | zoom | residual.